

1. Alice rolls a 6-sided dice until her dice shows a value of “3” for the first time, and then she stops. Similarly, Bob rolls a 20-sided dice until his dice shows a value of “3” for the first time, and then he stops.

Find the probability that each of them roll their respective dice the same number of times.

2. Consider two independent Geometric random variables, X and Y . Suppose that X has probability mass function $p_X(x) = P(X = x) = (3/5)^{x-1}(2/5)$ for integers $x \geq 1$. Suppose that Y has probability mass function $p_Y(y) = P(Y = y) = (4/9)^{y-1}(5/9)$ for integers $y \geq 1$. Calculate $P(Y \geq X)$.

3. Consider a 6-sided dice with 1 red side, 2 blue sides, and 3 yellow sides. Catherine rolls the dice repeatedly, until the red side appears, and then she stops. Let X denote the number of blue results during this process. Find the expected value of X .

4. Consider the independent random variables X and Y from Question 2. Find $P(X \geq 7 \mid Y \geq X)$.

5. Let X , Y , and Z be independent Geometric random variables that each have expected value $1/p$, i.e., $p_X(x) = (1-p)^{x-1}p$ for integers $x \geq 1$, and $p_Y(y) = (1-p)^{y-1}p$ for integers $y \geq 1$, and $p_Z(z) = (1-p)^{z-1}p$ for integers $z \geq 1$.

5a. Find $P(X + Y + Z > 10)$ in terms of p .

(Hint: Think about $X + Y + Z$ as the number of consecutive trials until the third success. We have $X + Y + Z > 10$ if and only if the first 10 trials have either 0 successes, 1 success, or 2 successes. In other words, $X + Y + Z \leq 10$ if and only if the 3rd success occurs within the first 10 trials.)

5b. Fix *any integer value of* $k \geq 2$. Find $P(X + Y + Z > k)$ in terms of p and k .